

Modelling the Spread of *Acinetobacter baumannii* in an Intensive Care Unit

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Motivation

- Carbapenem resistant *Acinetobacter baumannii* (CRAB) is a major threat to global public health, particularly in intensive care units (ICUs).
- Understanding how CRAB is transmitted between patients and the hospital environment is key to preventing disease spread.
- Data have kindly been provided by Willem van Schaik and colleagues in the Institute of Microbiology and Infection, University of Birmingham.
- We present a model based on regular longitudinal testing data of both patients and the bed units (including beds, sinks, and medical equipment) specific to each patient.
- We fit this in a Bayesian framework using the individual forward filtering backward sampling (iFFBS) algorithm.
- This demonstrates the utility of the iFFBS algorithm in fitting epidemic models at the individual level.

Colonisation Rates in the ICU

We consider the spread of bacteria between patients and bed units in the same room. Since we have testing data for the bed units as well as the patients, we introduce the bed units as “pseudo-individuals” with different colonisation rates to the patients.

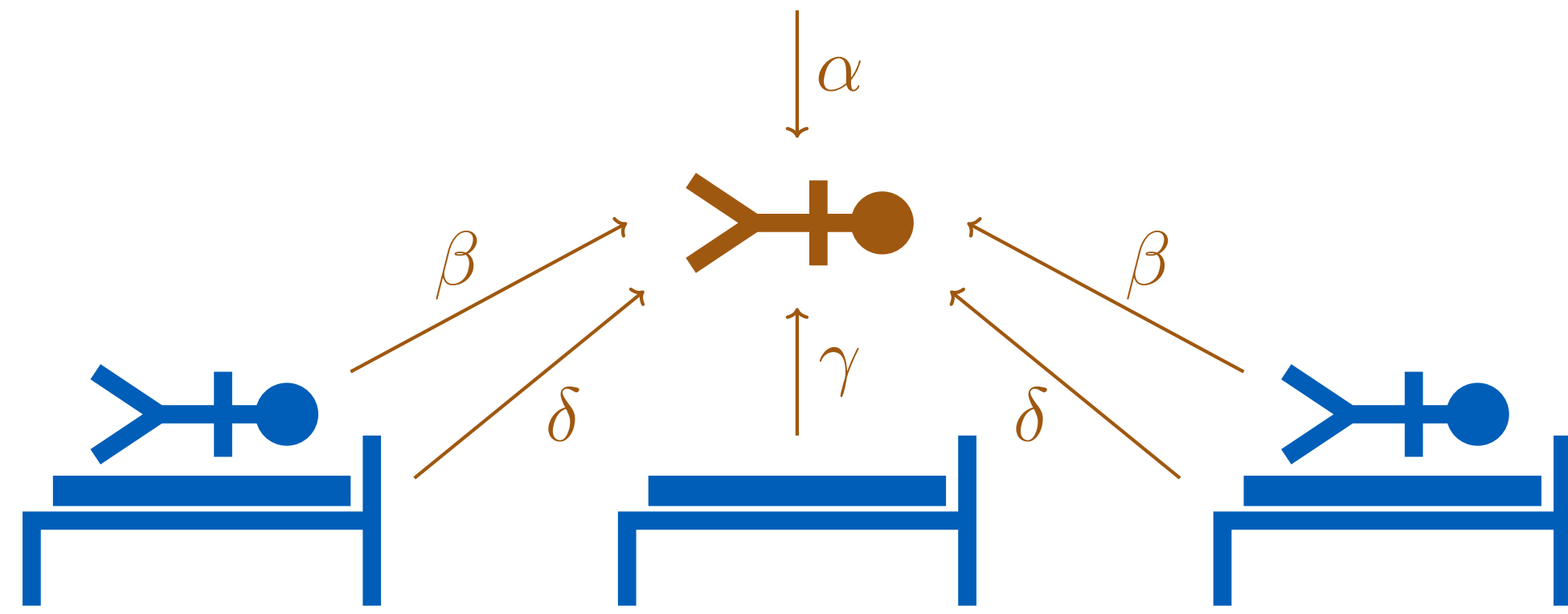


Figure 1. A patient in the ICU is colonised by: external sources at rate α ; other patients at rate β ; the bed unit occupied by the patient at rate γ ; and other bed units at rate δ .

We use the colonisation rates to derive the probability of patient p avoiding infection from day t to day $t + 1$ as

$$q_t^{(p)} = \exp \left(-\alpha - \beta P_t^{(-p)} - \gamma B_t^{(p)} - \delta B_t^{(-p)} \right),$$

where $P_t^{(-p)}$ is the number of colonised patients in the same room as patient p on day t ; $B_t^{(p)}$ is the colonisation status of the bed unit occupied by patient p on day t ; and $B_t^{(-p)}$ is the number of colonised bed units in the same room as patient p on day t .

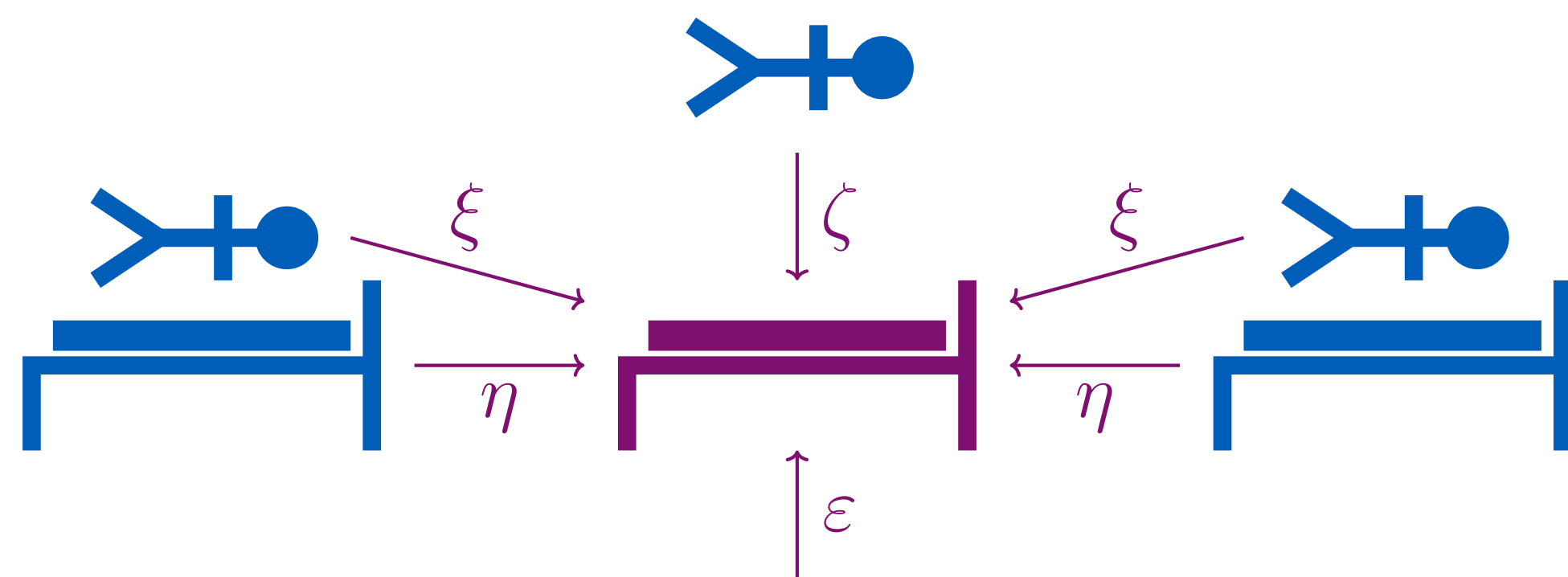


Figure 2. A bed unit in the ICU is colonised by: external sources at rate ϵ ; the patient occupying it at rate ζ ; patients in other bed units at rate ξ ; and other bed units at rate η .

We use the colonisation rates to derive the probability of bed unit b avoiding infection from day t to day $t + 1$ as

$$q_t^{(b)} = \exp \left(-\epsilon - \xi P_t^{(-b)} - \zeta P_t^{(b)} - \eta B_t^{(-b)} \right),$$

where $P_t^{(-b)}$ is the number of colonised patients in the same room as bed unit b on day t ; $P_t^{(b)}$ is the colonisation status of the patient occupying bed unit b on day t (if there is a patient in bed b); and $B_t^{(-b)}$ is the number of colonised bed units in the same room as patient p on day t .

Hidden Markov Models for Household Epidemic Data

We use a Coupled Hidden Markov Model (CHMM) to model the spread of CRAB in the ICU.

- The hidden states are the true colonization status of the patients and the bed units, modelled daily. We denote these with the matrices $\mathbf{P}_{1:T}^{1:N_p}$ and $\mathbf{B}_{1:T}^{1:N_b}$ respectively.
- The observations are the diagnostic test results for the patients and the bed units, measured weekly. We denote these with the matrices $\mathbf{X}_{1:T}^{1:N_p}$ and $\mathbf{Y}_{1:T}^{1:N_b}$ respectively.
- Patients and bed units are initially colonised with probability $\nu^{(p)}$ and $\nu^{(b)}$ respectively.
- The mean colonisation period for patients and bed units is m and r respectively.
- The sensitivity of the diagnostic test on patients and bed units is $\theta^{(p)}$ and $\theta^{(b)}$ respectively.

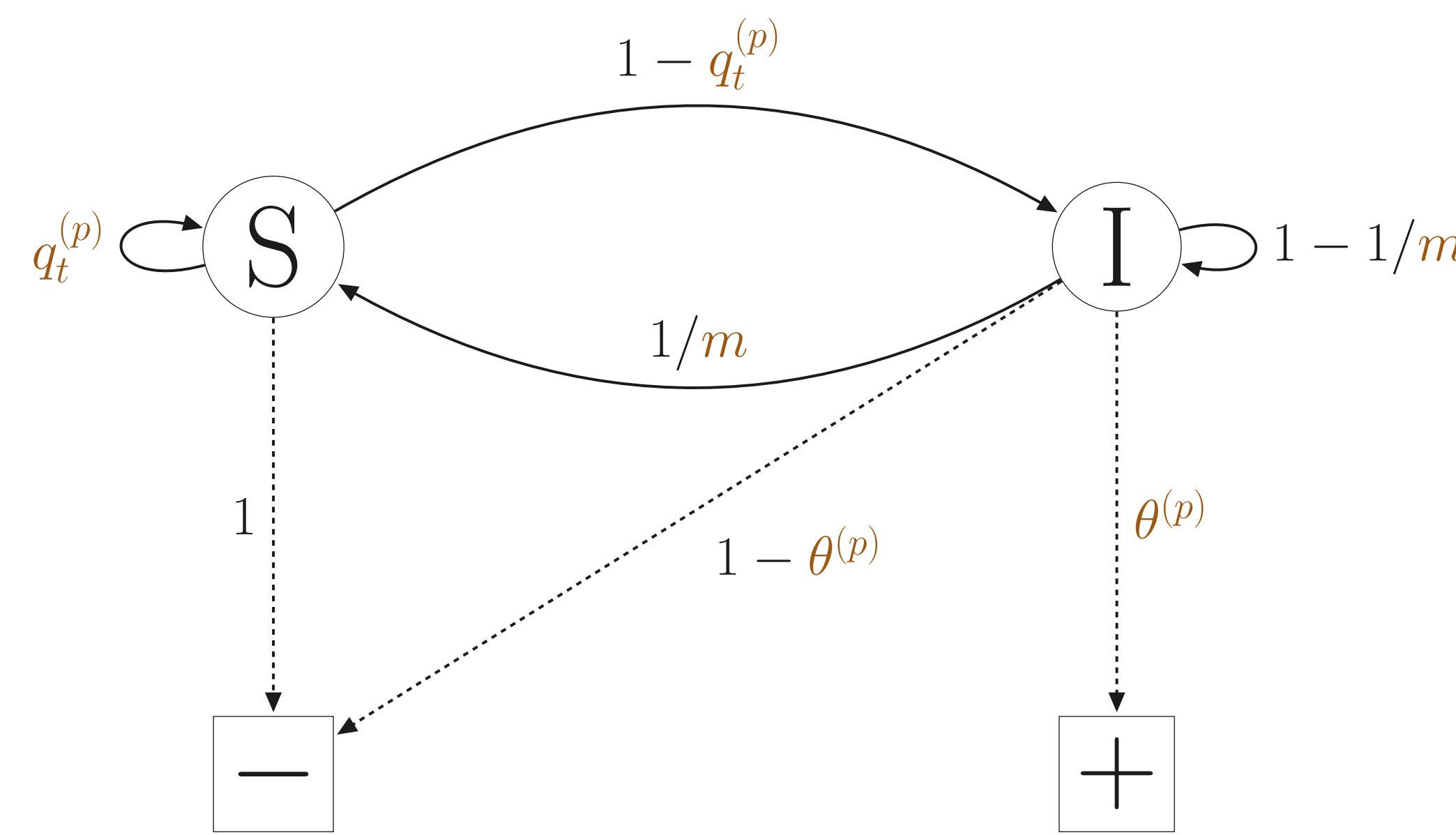


Figure 3. Hidden Markov Model for a patient in the ICU. Solid lines indicate transition probabilities between carriage states; dashed lines indicate observation probabilities given carriage states.

Parameter Inference

Given the observed data, we augment the model with $\mathbf{P}_{1:T}^{1:N_p}$ and $\mathbf{B}_{1:T}^{1:N_b}$ as additional parameters to be estimated. The iFFBS algorithm allows us to sample

$$\mathbf{P}_{1:T}^{(p)} \sim \pi \left(\mathbf{P}_{1:T}^{(p)} | \mathbf{X}_{1:T}^{(p)}, \mathbf{P}_{1:T}^{(-p)}, \mathbf{B}_{1:T}^{1:N_b}, \boldsymbol{\varphi} \right)$$

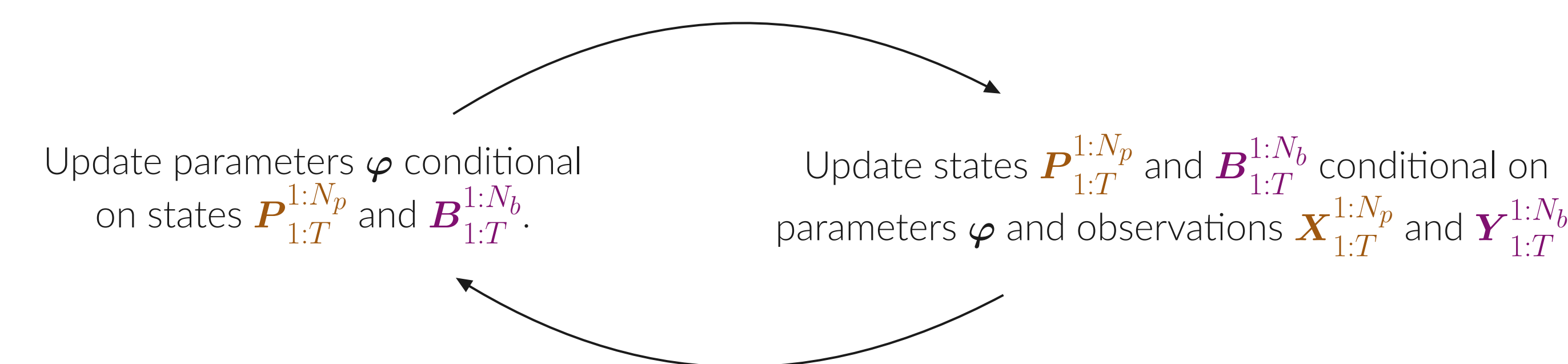
to update the hidden states for patient p , and

$$\mathbf{B}_{1:T}^{(b)} \sim \pi \left(\mathbf{B}_{1:T}^{(b)} | \mathbf{Y}_{1:T}^{(b)}, \mathbf{B}_{1:T}^{(-b)}, \mathbf{P}_{1:T}^{1:N_p}, \boldsymbol{\varphi} \right)$$

to update the hidden states for bed unit b , where $\boldsymbol{\varphi}$ is the model parameters. We repeat this for $p = 1, \dots, N_p$ and $b = 1, \dots, N_b$, which generates a new sample of $\mathbf{P}_{1:T}^{1:N_p}$ and $\mathbf{B}_{1:T}^{1:N_b}$.

We update the parameters $\boldsymbol{\varphi}$ using a hybrid Gibbs sampler:

- Parameters $(\alpha, \beta, \gamma, \delta)$ are updated with a Hamiltonian step.
- Parameters $(\epsilon, \xi, \zeta, \eta)$ are updated with a Hamiltonian step.
- Parameters $(m, r, \nu^{(p)}, \nu^{(b)}, \theta^{(p)}, \theta^{(b)})$ are updated with individual Gibbs updates from their marginal posterior distributions.



Simulation Study

We ran a simulation study informed by the real dataset which we are using to develop this model.

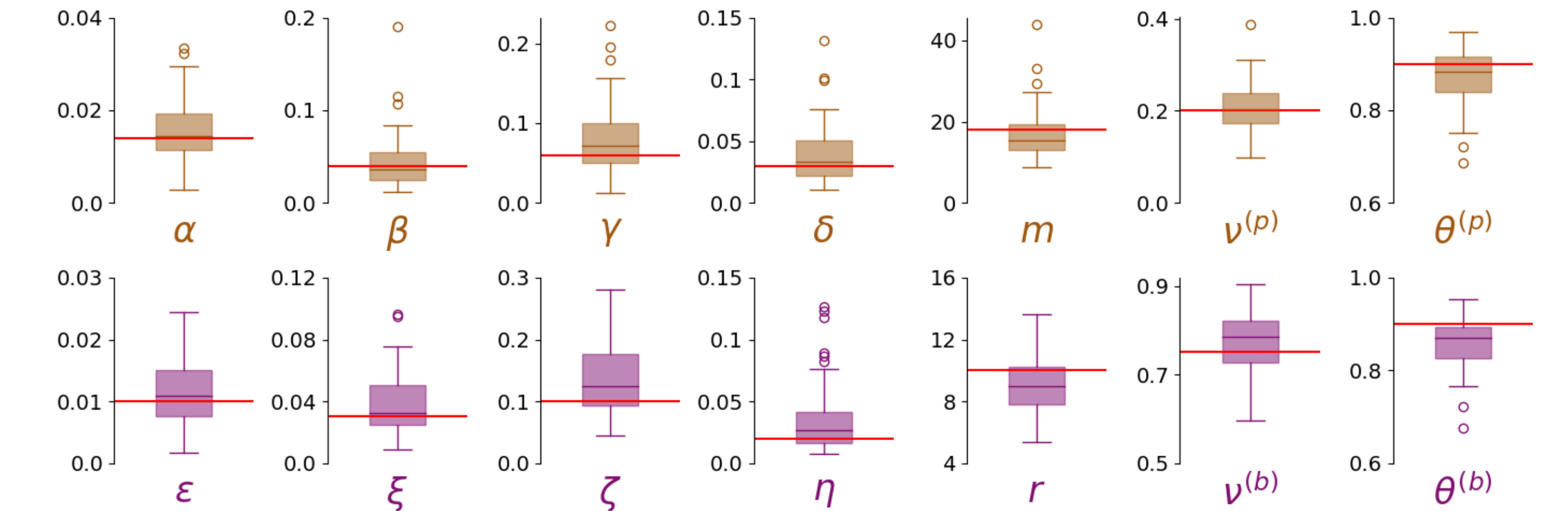


Figure 4. Boxplots of estimated posterior medians, with the true value of the parameter shown as a horizontal line.

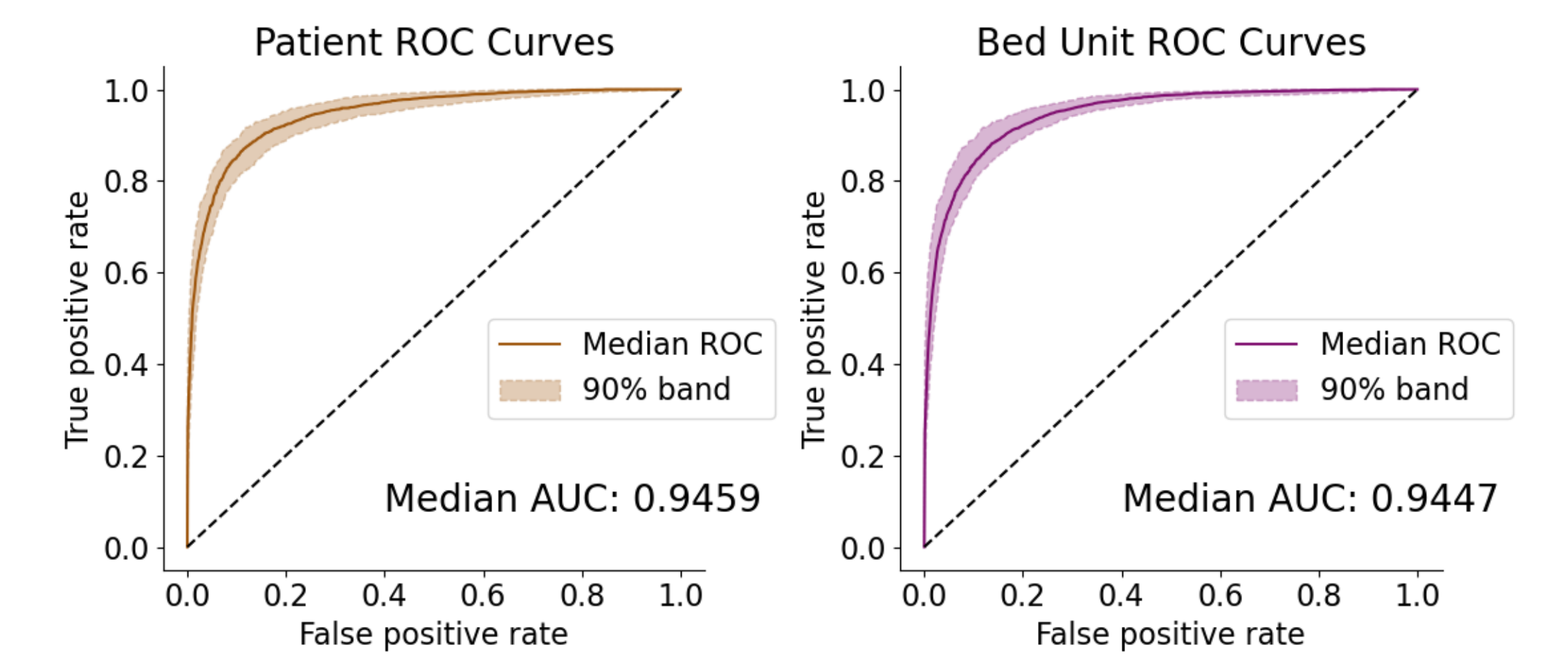


Figure 5. Median receiver operating characteristic (ROC) curve and 90% quantile band from the simulation study.

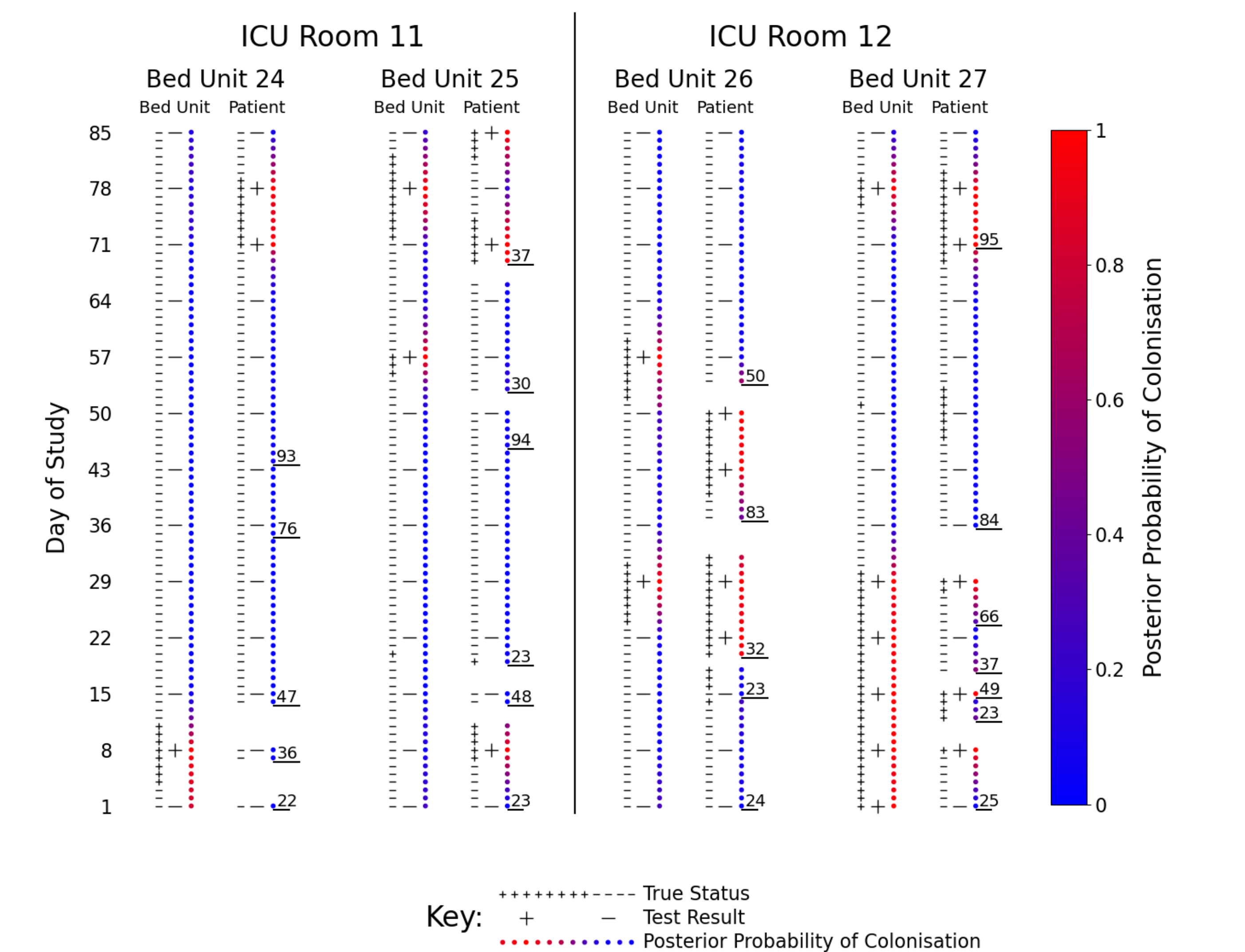


Figure 6. Simulated data for Rooms 11 and 12, showing the spread of CRAB as patients move between bed units.